



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

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Division: A1

ASSIGNMENT No: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Outcome:

Interfaces were successfully created and implemented across multiple, unrelated classes, with each class providing its own implementation of the same abstract method(s), as verified through the sample programs.

```
interface Drawable{
    void draw();

    double PI = 3.14159;
}

class Square implements Drawable{
    double side;

    Square(double s){
        this.side = s;
    }

    public void draw(){
        System.out.println("Drawing a Sqaure with side " + side);
    }
}

class Circle implements Drawable{
    double radius;

    Circle(double r){
        this.radius = r;
    }

    public void draw(){
        System.out.println("Drawing a Circle with radius " + radius);
        System.out.println("Area using interface contstant PI: " + (PI * radius *
radius));
    }
}

public class JAVA_EXP_11{
    public static void main(String [] args){

        Drawable shape1 = new Square(5);
        shape1.draw();

        System.out.println();

        Drawable shape2 = new Circle(4);
        shape2.draw();
    }
}
```

Output:

```
in' 'JAVA_EXP_11'  
Drawing a Sqaure with side 5.0  
  
Drawing a Circle with radius 4.0  
Area using interface contstant PI: 50.26544  
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

Conclusion:

Therefore, from the experiment we got to learn the implementation of interference.

Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

Source Code:

```
1  interface Printable {
2      void printDetails();
3  }
4
5  class Student implements Printable {
6      int rollNo;
7      String name;
8
9      Student(int rollNo, String name) {
10         this.rollNo = rollNo;
11         this.name = name;
12     }
13
14     public void printDetails() {
15         System.out.println(x: "Student Details");
16         System.out.println("Roll No : " + rollNo);
17         System.out.println("Name      : " + name);
18     }
19 }
20
21 class Employee implements Printable {
22     int empId;
23     String empName;
24
25     Employee(int empId, String empName) {
26         this.empId = empId;
27         this.empName = empName;
28     }
29
30     public void printDetails() {
31         System.out.println(x: "Employee Details");
32         System.out.println("Employee ID : " + empId);
33         System.out.println("Name          : " + empName);
34     }
35 }
36
37 public class assignment11_PrintableDemo {
38     Run | Debug | Run main | Debug main
39     public static void main(String[] args) {
```

```

38     public static void main(String[] args) {
39
40         Student s = new Student(rollNo: 101, name: "Aashray");
41         Employee e = new Employee(empId: 201, empName: "Aarya");
42
43         s.printDetails();
44         System.out.println();
45         e.printDetails();
46     }
47 }

```

Output:

```

C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment11_PrintableDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Student Details
Roll No : 101
Name    : Aashray

Employee Details
Employee ID : 201
Name       : Aarya

```

2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

Source Code:

```
1  interface Switchable {
2      void turnOn();
3  }
4
5  class Light implements Switchable {
6
7      public void turnOn() {
8          System.out.println(x: "Light is ON.");
9      }
10 }
11
12 class Fan implements Switchable {
13
14     public void turnOn() {
15         System.out.println(x: "Fan is ON.");
16     }
17 }
18
19 public class assignment11_SwitchDemo {
20     Run | Debug | Run main | Debug main
21     public static void main(String[] args) {
22
23         Light light = new Light();
24         Fan fan = new Fan();
25
26         light.turnOn();
27         fan.turnOn();
28     }
29 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment11_SwitchDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Light is ON.
Fan is ON.
```